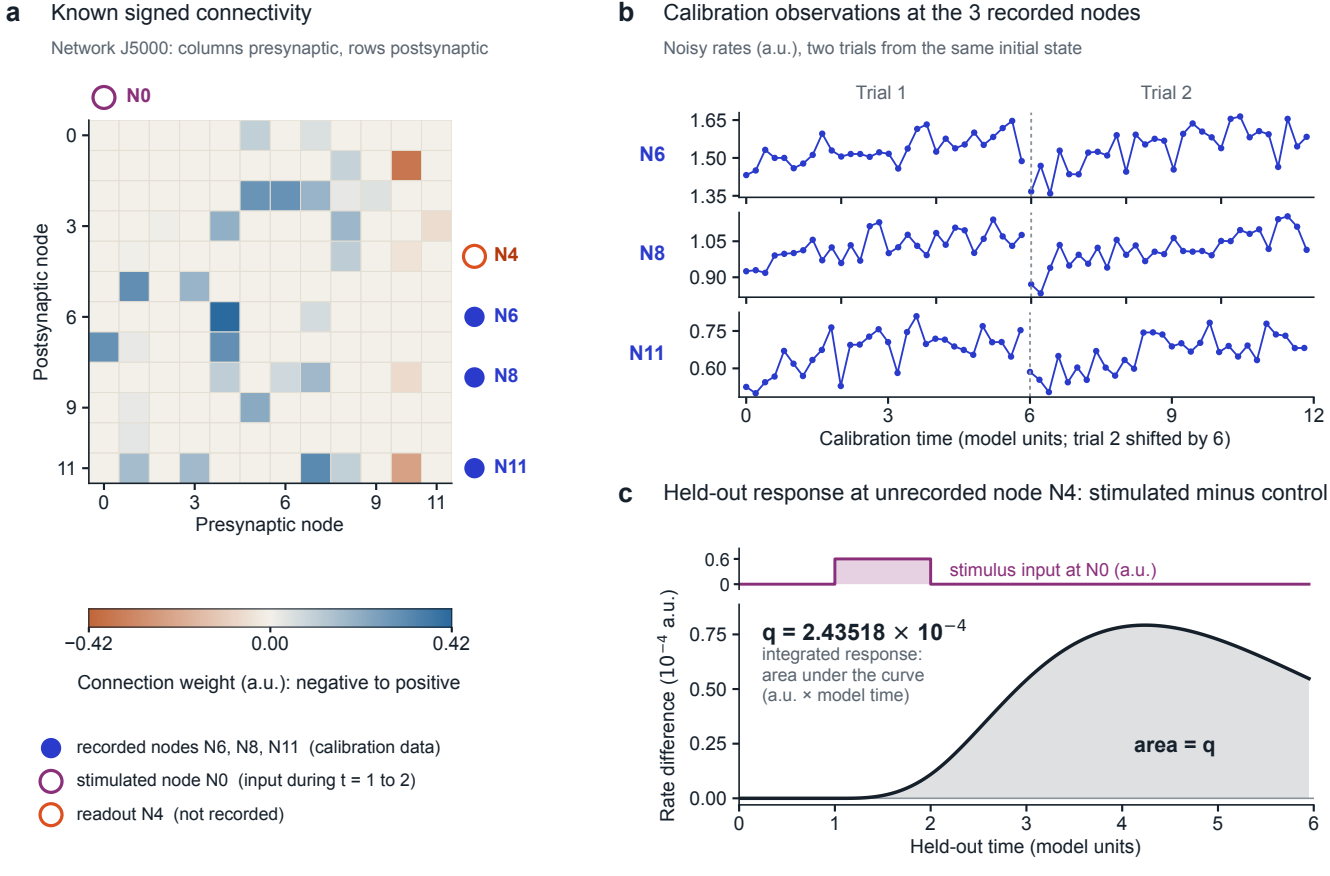

a
Known signed connectivity
Network J5000: columns presynaptic, rows postsynaptic
N0
N4
N6
N8
N11
Postsynaptic node
Presynaptic node
0
3
6
9
11
−0.42
0.00
0.42
Connection weight (a.u.): negative to positive
recorded nodes N6, N8, N11 (calibration data)
stimulated node N0 (input during t = 1 to 2)
readout N4 (not recorded)
b
Calibration observations at the 3 recorded nodes
Noisy rates (a.u.), two trials from the same initial state
Trial 1
Trial 2
N6
N8
N11
1.65
1.50
1.35
1.05
0.90
0.75
0.60
0
3
6
9
12
Calibration time (model units; trial 2 shifted by 6)
c
Held-out response at unrecorded node N4: stimulated minus control
stimulus input at N0 (a.u.)
0.6
0
q = 2.43518 × 10−4
integrated response:
area under the curve
(a.u. × model time)
area = q
Rate difference (10−4 a.u.)
0.75
0.50
0.25
0.00
0
1
2
3
4
5
6
Held-out time (model units)